

FEDERATED DUAL-MODEL FRAMEWORK FOR PRIVACY-PRESERVING LEGAL CONTRACT ANALYSIS

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Abstract—Legal contract analysis can be one of many ways to use Natural Language Processing (NLP) as companies have contracts in large volumes that include obligations and liability. Most NLP systems based upon Machine Learning are currently trained centrally requiring all the data used to train the system to be put into a single location. This creates a huge risk for privacy when using legal contracts since they contain sensitive information about the parties involved. In response to the issues identified above, we propose to create a Hybrid Federated Learning Architecture that utilizes the LegalBERT Model for Clause Classification and the Fine-Tuned Phi-2 Language Model that has been enhanced utilizing LoRA Adapters for Explanation and Summarization. The proposed system will utilize the Federated Averaging Algorithm allowing for Collaborative Learning across Distributed Clients while preventing the need to share the raw contractual data. Experiments on the LEDGAR dataset achieve 0.85 classification accuracy under federated conditions compared to 0.88 in centralized training, with the difference attributed to non-uniform data distribution across clients, and generates summaries with ROUGE-1, ROUGE-2, and ROUGE-L scores of 0.57, 0.32, and 0.57. The results show that accurate and scalable contract analysis can be achieved while still preserving data privacy.

Keywords— Federated Learning, Machine Learning, Legal Contract Analysis, Transformer Models, Distributed AI, Privacy Preservation.

I. INTRODUCTION

Contract review is a critical legal process involving analysis of agreements to ensure compliance with regulations, clause classification, and anomaly detection. Modern organizations handle tens of thousands of contracts and manually reviewing them is time-consuming and costly due to high dimensionality and the administrative burden [1]. Studies report over half of contract review tasks such as identifying and classifying clauses are repetitive and demand specialized legal expertise. Natural Language Processing (NLP) and Machine Learning (ML) methods have therefore become central to automating contract analysis. Legal texts are typically unstructured, free-form documents, but NLP methods can convert such text into structured features for automated analysis. Especially, ML-driven Technology Assisted Review tools use text classification, context preservation and transformer-based

models to locate relevant documents, extract clauses and flag potential compliance issues in specialised domain's [2].

That said, applying standard Artificial Intelligence (AI) techniques to contracts can lead to rise in data privacy and security challenges. Contracts contain sensitive details like client information, financial details and trade secrets. ML approaches operate on aggregating all contract data into one repository, which risks data breaches and can violate regulations and compliance with GDPR or confidentiality requirements [3]. For example, direct sharing of sensitive data is often impossible under strict privacy laws without methods like Federated Learning for improved collaboration but still preserving privacy [4]. So, to address this, federated learning has emerged as a promising solution. It enables collaborative model training across multiple institutions without moving the raw contract data. By keeping legal documents on local servers and sharing only model updates, federated learning preserves data confidentiality.

In this work, we present a hybrid federated learning system for contract review that combines a domain-specific decoder model and a generative model. We use a pretrained Legal-BERT encoder to classify clauses, and a compact Transformer (LoRA-adapted) decoder model to generate explanations and summaries of contract text. Training is performed under a federated approach, so that each node trains locally on its own contract data. The server aggregates only local encrypted model weights, ensuring no private legal text is exchanged. This architecture allows joint training of large language models while meeting strict privacy requirements [5]. Our approach is evaluated on the LEDGAR contract dataset, demonstrating that clause classification accuracy and summarization quality remain high even under federated training. In short, our system shows distributed AI can enable automated contract analysis across organizations without compromising data privacy.

II. RELATED WORK

Recent related work and improvements in Natural Language Processing (NLP) and Machine Learning (ML) have made it feasible for the automated analysis of legal documents. This analysis includes classification and summarization of clauses,

anomaly detection, and contractual risk identification. At the same time, concerns about the privacy of legal data have led to the adoption of privacy-preserving learning approaches, especially federated learning. Similarly, transformer-based language models have improved, now being able to capture contextual semantics in complex legal text. The following works demonstrate how these changes contributed to decentralized learning, legal NLP, and efficient model adaptation.

Sonani et al. [5] presented a federated learning-based framework for contract review, where decentralized training is performed with Differential Privacy and Secure Multi-Party Computation to ensure data confidentiality. Their findings demonstrate that federated learning facilitates secure collaboration among legal entities while still preserving performance comparable to centralized models, illustrating that contract analysis tasks such as clause classification and anomaly detection can be executed without the necessity of sharing raw data. The paper additionally discusses challenges that arise when deploying federated architectures in the real world, such as data heterogeneity and computational complexity. This shows that federated architectures need to be efficient and scalable.

Dechtiar et al. [6] proposed a transformer-based approach for detecting contract smells, using BERT and LegalBERT models to identify inconsistencies in legal contract drafting. The research illustrates that pretrained transformer models can effectively identify semantic irregularities and contextual dependencies within contractual clauses. The results also show that contextual embeddings can reliably find structural inconsistencies. These findings demonstrate that transformer models can understand complex patterns in legal language.

Dikmen et al. [7] developed an NLP-based system for analysing construction contracts, focusing on clause extraction and classification of contractual risks and responsibilities. Their findings indicate that ML models can effectively detect risk-related clauses and contractual obligations, thereby supporting automated contract interpretation. This demonstrates the importance of structured text analysis for decision-making in specialized legal domains.

Piccialli et al. [8] conducted an extensive study of federated and edge learning for Large Language Models (LLMs), evaluating distributed optimization, communication efficiency and privacy mechanisms. The survey states that federated learning environments can assist with scalable model training while still preserving privacy. It also discusses challenges such as communication overhead, model convergence, and heterogeneous environments and data distributions that affect the performance of distributed systems.

Wong et al. [9] introduced a knowledge-augmented transformer framework for construction contract risk identification, integrating domain-specific expertise into pretrained language models. The results indicate that

incorporating vectorized domain expert knowledge improves contextual understanding and interpretability of contract analysis, enabling more accurate identification of risks and contractual implications.

Zhang et al. [10] proposed an efficient federated optimization method (FedFA) for LLMs, addressing issues related to non-IID (non-independent and identically distributed) data and client heterogeneity. Their findings show that adaptive aggregation strategies improve training stability and managing disparities in distributed environments, highlighting the importance of optimization techniques in federated learning systems.

Wang et al. [11] presented a parameter-efficient knowledge distillation adaptation method for pretrained language models using adapter-based architectures. The study demonstrates that low-rank adaptation significantly reduces computational overhead while maintaining model performance on downstream tasks, enabling efficient fine-tuning of LLMs in practical scenarios.

Collectively, these works indicate that federated learning enables privacy-preserving collaboration, transformer-based models provide effective semantic understanding of legal text, and parameter-efficient techniques support scalable adaptation of LLMs, forming the basis for integrated approaches to automated contract analysis. The following section explains our proposed dual-model architecture for privacy-preserving legal contract analysis.

III. PROPOSED METHODOLOGY

The proposed system is evaluated using the LEDGAR dataset, a large-scale corpus particularly developed for multi-label classification of legal contractual provisions. The dataset is constructed from contracts filed with the Securities and Exchange Commission (SEC) and contains over 60,000 clauses annotated across 98 semantic categories, where each clause is assigned labels corresponding to its legal function such as governing law, indemnification and termination [12]. This formulation reflects real-world contract review settings where clauses are mapped to functional categories for automated analysis.

Previous frameworks on LEDGAR highlight the dataset exhibits a long-tail label distribution, where a small subset of clause categories dominates the corpus while many labels occur infrequently. To address this imbalance and improve training stability, we restrict the classification space to the Top-K most frequent labels ($K = 20$), a commonly adopted approach in legal NLP to handle sparsity and improve generalization in multi-label settings.

The dataset is then distributed across multiple federated clients, where each client receives a non-independent and non-identically distributed (non-IID) subset of clauses. This partitioning strategy reflects real-world conditions in which legal data is inherently decentralized across organizations with varying contract types and clause

distributions [13]. Such heterogeneity has been identified as a key challenge in federated learning systems, requiring the need of a universal heterogenous benchmark dataset for federated learning and influencing model convergence and performance.

Before training, the clauses undergo preprocessing, which is consistent with established NLP pipelines for legal text. The text is normalized through lowercasing and removal of non-informative tokens, preserving domain-specific legal terminology. Tokenization is performed using subword-based methods such as WordPiece, which has shown improvements in the handling of rare and domain-specific terms in transformer-based architectures. Legal documents are characterized by complex sentence structures, nested conditions and long contextual dependencies, which make sequence modeling particularly challenging.

The dataset analysis in Fig.1 reveals a skewed distribution of clause categories, with frequent clauses like governing law and notices appearing in a significant portion of contracts. This imbalance is a result of common drafting practices and has an effect on how effectively minority classes are classified.

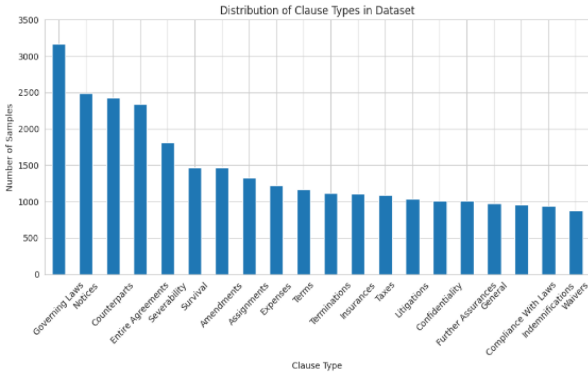


Fig. 1 Distribution of clause types in the LEDGAR Dataset

Clause length statistics represented in Fig.2 further indicate significant variability, where many clauses are moderately sized while others extend to long sequences due to embedded legal conditions. Such variability necessitates models capable of capturing long-range dependencies in text.

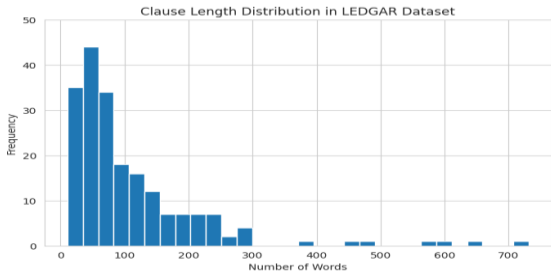


Fig. 2 Clause Length Distribution in LEDGAR Dataset

Vocabulary analysis in Fig.3 and Fig.4 reveals that legal word embeddings on large corpora contain repetitive domain-specific terminology, including terms such as

agreement, party, and liability, which contribute to stable contextual embeddings in pretrained models and are of major importance in NLP [14].



Fig. 3 Corpora of frequent Legal terms in LEDGAR Dataset

The prevalence of particular clause types offers robust supervision signals during training, whereas infrequent categories introduce semantic diversity in classification tasks. The label imbalance, heterogeneous clause structure, and domain-specific vocabulary in these datasets are well-known problems in legal NLP that make transformer-based domain-adapted models necessary.

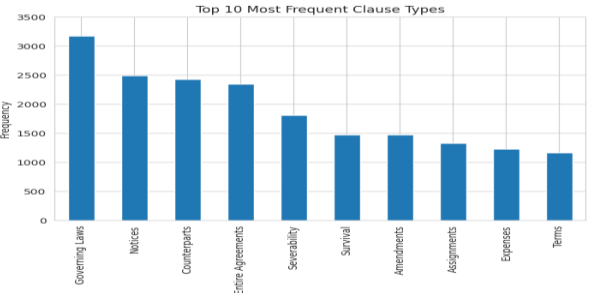


Fig. 4 Top 10 Most Frequent Clause types in LEDGAR Dataset

Based on these observations, the proposed system adopts a hybrid dual-model architecture integrating a BERT neural model [15] based LegalBERT encoder for clause classification, and a parameter-efficient generative transformer model for explanation and summarization, both trained within a federated learning framework to ensure decentralized data processing while enabling collaborative model optimization, with LEGALBERT belonging to the family of the BERT models with assistance for legal NLP research [16].

The procedural steps for federated training of the clause classification component of the LegalBERT encoder model are outlined below in Algorithm 1. The training follows a decentralized optimization process in which each client updates the model locally using its private dataset, and only model parameters are shared with the central server, consistent with federated learning principles.

Algorithm 1: Federated encoder model Clause Classification Training

Input:

Input:

 K : Total number of clients

 D_i : Local dataset at client i , where $i \in \{1, 2, \dots, K\}$
 n_i : Number of samples in D_i
 N : Total number of samples across all clients ($N = \sum_{i=1}^K n_i$)

 W_t : Global LegalBERT encoder parameters of communication round t
 η : Learning rate

 T : Total number of communication rounds

 $L_i(W)$: Loss function computed on dataset D_i of weight W
Output:
 W_T : Final trained global encoder model parameter

- 1: Initialize global model parameters W_0 at the server
 - 2: **for** each communication round $t = 0$ to $T-1$ **do**
 - 3: Broadcast current global model W_t to all clients
 - 4: **for** each client $i \in \{1, 2, \dots, K\}$ in parallel **do**
 - 5: Load local dataset D_i
 - 6: Compute local gradient $\nabla L_i(W_t)$
 - 7: Update local model parameters
using $W_{t+1}^i = W_t - \eta \nabla L_i(W_t)$
 - 8: **end for**
 - 9: Aggregate local models at server using Federated Averaging:

$$W_T = \sum_{i=1}^K (n_i/N) W_{t+1}^i$$
 - 10: **end for**
 - 11: Return final global model W_T
 - 12: **end**
-

While the encoder model is responsible for clause classification, the generative decoder component is optimized using parameter-efficient adaptation techniques. Instead of updating the full set of parameters in the Phi-2 model, only the LoRA adapter parameters are trained, while the base model weights remain fixed. This approach reduces computational overhead and communication cost, which is critical for federated environments involving LLMs. Each client performs local training of adapter parameters using its private dataset and transmits only the adapter updates to the central server. The aggregation mechanism follows the same weighted averaging strategy but operates exclusively on adapter parameters, aligning with recent approaches for efficient federated training of large models. The federated adapter training procedure for

the generative decoder component is described below in Algorithm 2.

Algorithm 2: Federated LoRA Adapter Training for generative decoder model

Input:
 K : Total number of clients

 D_i : Local dataset at client i , where $i \in \{1, 2, \dots, K\}$
 n_i : Number of samples in D_i
 N : Total number of samples across all clients ($N = \sum_{i=1}^K n_i$)

 A_t : Global LoRA adapter parameters at communication round t
 W_{base} : Pretrained Phi-2 base decoder model parameters (frozen)

 η : Learning rate

 T : Total number of communication rounds

 $L_i(A)$: Loss function computed on dataset D_i for LoRA parameter A
Output:
 A_T : Final trained global LoRA adapter parameter

- 1: Initialize global adapter parameter A_0 at the central server
 - 2: **for** each communication round $t = 0$ to $T-1$ **do**
 - 3: Broadcast adapter parameters A_t to all clients
 - 4: **for** each client $i \in \{1, 2, \dots, K\}$ in parallel **do**
 - 5: Load local dataset D_i
 - 6: Freeze base model parameters W_{base}
 - 7: Compute local gradient on dataset D_i : $\nabla L_i(A_t)$
= gradient of L_i with respect to A_t
 - 8: Update local adapter parameters:
 $A_{t+1}^i = A_t - \eta \nabla L_i(A_t)$
 - 9: **end for**
 - 10: Aggregate local adapter parameters using Federated Averaging: $A_T = \sum_{i=1}^K (n_i/N) A_{t+1}^i$
 - 11: **end for**
 - 12: Return final global adapter parameter A_T
 - 13: **end**
-

In this federated optimization process, the clause classification encoder and the generative reasoning decoder are trained together in different places while ensuring that each client's raw contractual data remains localized at each client. The complete architecture of the proposed hybrid system is shown in Fig.5.

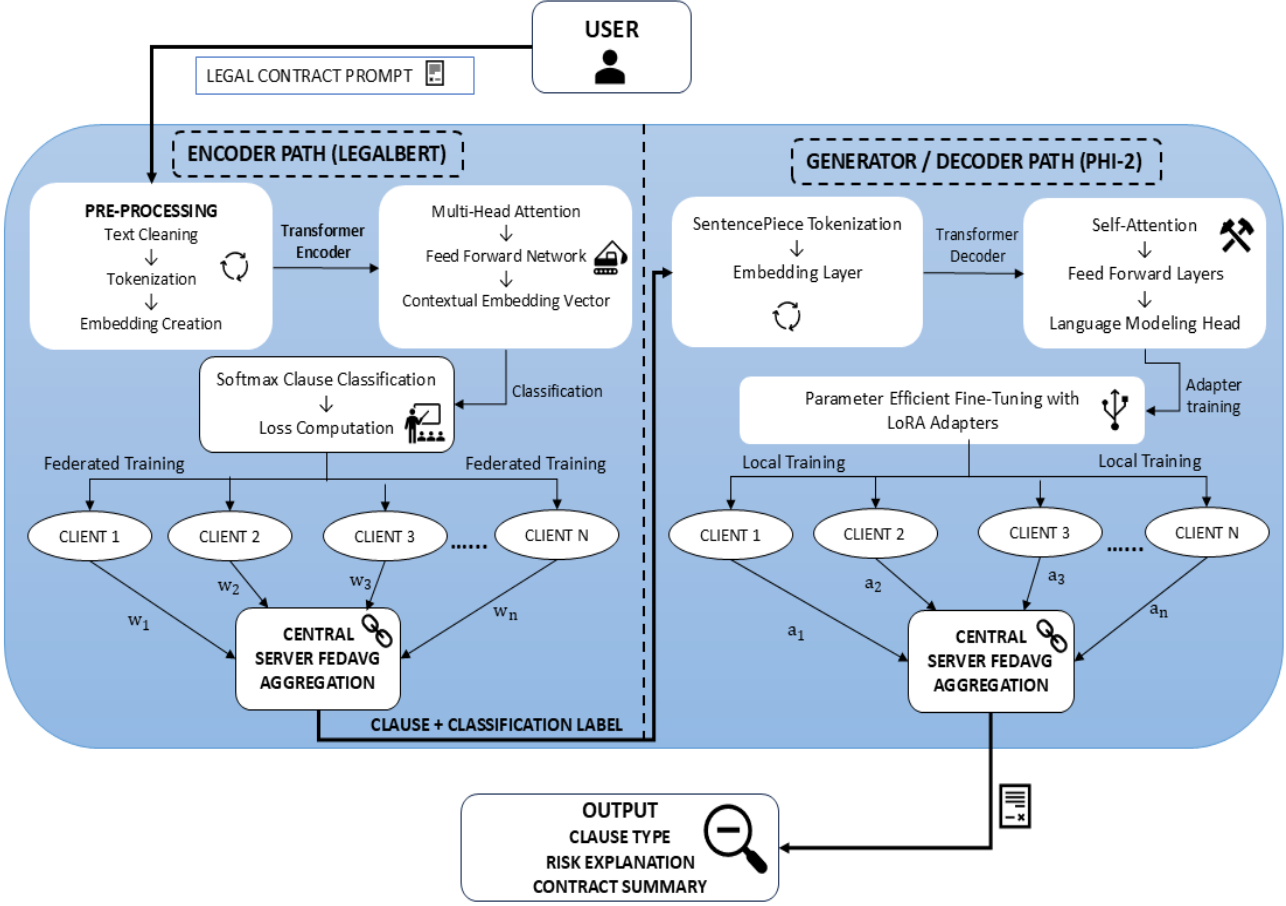


Fig. 5 Complete Proposed Hybrid Architecture

The workflow begins when the user provides a legal contract or text prompt as input. The text first goes through preprocessing steps including text normalization, token segmentation, and embedding construction. Text normalization removes formatting symbols that are irrelevant while retaining legal terms which are specific to the domain. The HuggingFace tokenizer which is included with the LegalBERT model uses the WordPiece tokenization algorithm to separate all tokens for the encoder part to store vocabulary token-to-index mapping for the encoder model [17]. WordPiece tokenization breaks down long words into smaller parts called "subwords" to improve vocabulary coverage and reduce down on tokens which aren't in the vocabulary.

The following equation 1 provides a given input clause represented as a token sequence

$$\mathcal{X} = x_1, x_2, \dots, x_n \quad (1)$$

where each token x_n is mapped into a dense vector representation through an embedding layer for total n tokens. The embedding representation $H^{(0)}$ combines token embeddings, positional embeddings, and segment embeddings and is computed in equation 2

$$H^{(0)} = E(x_i) + P(x_i) + S(x_i) \quad (2)$$

where $E(x_i)$ represents the token embedding for token x_i , $P(x_i)$ represents positional encoding that captures the sequential order of tokens and $S(x_i)$ represents segment embeddings used for contextual differentiation.

The transformer encoder architecture of the LegalBERT model then processes the embedded token representations. The encoder is made up of several transformer layers that are stacked on top of each other. These layers use the self-attention mechanism to model how tokens relate to each other. Within each transformer layer, the input representation is projected into query (Q), key (K), and value (V) vectors defined in equations 3,4,5

$$Q = X W_Q \quad (3)$$

$$K = X W_K \quad (4)$$

$$V = X W_V \quad (5)$$

where X represents the input embedding matrix and W_Q , W_K , and W_V represent learnable projection matrices. Attention

weights between tokens are computed using scaled dot product attention using softmax classifier [18] defined in equation 6

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{(QK^T)}{\sqrt{d_k}}\right)V \quad (6)$$

where d_k denotes the dimensionality of the key vectors. This mechanism allows the model to understand how words in the clause are related to each other.

To improve representation capacity, the encoder employs multi head attention, which performs attention computation in multiple parallel subspaces. Multi head attention is defined in equation 7

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h)W_o \quad (7)$$

where each head performs an independent attention operation which is finally concatenated. The resulting representations are then passed through a position wise feed forward network (FFN) defined in equation 8

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2 \quad (8)$$

where $\max()$ refers to the maximum operation and b_1 and b_2 are learnable bias vectors.

After each attention and feed forward layer, residual connections and layer normalization are used to keep the gradient from getting too big during training. Once the transformer encoder has processed through the clause, it extracts the contextual embedding corresponding to the classification token and sends it to a classification layer. Clause classification is defined as a multi-class classification problem where the model predicts the probability distribution across clause categories. The predicted class probabilities are computed using a softmax classifier [19] defined in equation 9

$$P(y|x) = \text{softmax}(W \cdot h + b) \quad (9)$$

where h represents the contextual embedding generated by the encoder, W denotes the classifier weight matrix, and b represents the bias term.

The model parameters are optimized using the categorical cross entropy loss function using the following equation 10 [20]

$$L = -\sum y_i \log(p_i) \quad (10)$$

where y_i represents the true label and p_i represents the predicted probability. The AdamW optimization algorithm, which combines adaptive moment estimation with weight decay regularization [21] is used to optimize the model parameters. The AdamW optimizer updates parameters according to equation 11

$$\theta_{t+1} = \theta_t - \eta * \hat{m}_t / (\text{sqrt}(\hat{v}_t) + \varepsilon) \quad (11)$$

where η represents the learning rate, θ_t represents model parameter at step t , $\hat{m}_t$ and $\hat{v}_t$ denote bias corrected first and

second moment estimates of gradients, and ε is a small numerical stability constant to prevent division by zero.

The clause classification model is trained in a federated learning setup to keep legal documents confidential. In this case, the dataset is distributed across several simulated clients that are representative of separate legal entities such as law firms or corporate legal departments. Each client trains the model locally using its private dataset and only shares model updates with a central aggregation server. Let W_{t+1}^i represent the model parameters learned by client i with local dataset size n_i at communication round $t+1$. Local model updates are computed using stochastic gradient descent [22] given in equation 12

$$W_{t+1}^i = W_t - \eta \nabla L_i(W_t) \quad (12)$$

where $\nabla L_i(W^{(t)})$ is the gradient of the loss function L with respect to the model parameter W .

The locally updated parameters from each client are transmitted to the central server and aggregated using the Federated Averaging algorithm [23]. The global model parameters are updated according to the following equation 13

$$W_T = \sum_{i=1}^K (n_i/N) W_{t+1}^i \quad (13)$$

where $N = \sum_{k=1}^K n_k$ represents the total number of training samples across all participating clients K . The aggregated global model is then redistributed to clients to begin the next training round. This iterative procedure allows collaborative learning across organizations while ensuring that raw contractual data never leaves the local environment.

After clause classification is done, the predicted clause category and the original clause text are sent to the architecture's generative reasoning component. This component is built using the Phi-2 large language model which is based on a decoder-only transformer architecture designed for autoregressive language generation [24]. The generative model uses SentencePiece tokenization, which breaks text into subword units based on patterns of how often words appear in large corpora [25]. This is different from the encoder component.

The tokenized input is converted into embeddings and processed through stacked transformer decoder layers. Each decoder layer uses masked self-attention to make sure that token generation only depends on tokens that have already been generated. The probability of generating the next token x_t in a sequence is defined as

$$P(x_t | x_1, x_2, \dots, x_{t-1})$$

Due to the large parameter size of the generative model, full fine-tuning is computationally expensive. Therefore parameter-efficient fine-tuning is implemented using the Low Rank Adaptation (LoRA) method [26]. In this approach the

pretrained model weights remain frozen while low-rank adapter matrices are introduced to capture task-specific updates. The adapted weight matrix is computed in equation 14

$$W = W^0 + BA \quad (14)$$

where W^0 represents the original pretrained weight matrix and where $A \in R_{r \times d}$ and $B \in R_{r \times d}$ are low-rank matrices with rank $r \ll d$ significantly smaller than the original model dimensionality. This technique reduces the number of trainable parameters by a huge amount while preserving the base language model's expressive capacity. Similar to the encoder component, LoRA adapters are trained within the federated learning environment. Each client locally trains adapter parameters using its private dataset while keeping the base Phi-2 model weights fixed. Let A_t represent the adapter parameters learned by client i at communication round t . Adapter updates are aggregated at the central server after all communication rounds T using Federated Averaging given in equation 15

$$A_T = \sum_{i=1}^K (n_i/N) A_{t+1}^i \quad (15)$$

This adapter level federation significantly reduces communication overhead compared to transmitting full model parameters.

During the inference stage, a hybrid pipeline combines the outputs of both components. The clause classification model first predicts the clause category, and then uses that information to create a structured prompt for the generative model. Using autoregressive decoding, the generative model generates contextual explanations and short summaries of the contract clauses. To keep the text coherent and diverse, text generation uses probabilistic sampling methods such as temperature scaling and nucleus sampling. In the implemented system, temperature is set to 0.7, nucleus sampling threshold is set to 0.9, and generation length is limited to a maximum of 150 tokens, which can be modified according to user preference.

The final output generated by the system consists of three elements: the predicted clause category, a natural language explanation describing potential contractual risks or implications, and a concise summary of the clause content. By combining transformer-based clause classification with parameter-efficient federated large language models, the proposed architecture provides a scalable and privacy-preserving solution for automated legal contract analysis across distributed organizational environments.

To enable collaborative model optimization across distributed legal organizations, the training procedure of the classification and generative components is implemented using a federated learning framework. In this setting, the training dataset is partitioned across multiple simulated clients (nodes), each representing an independent legal repository like a law firm, regulatory body or corporate legal department. Each client performs model training locally using its private

contractual dataset while sharing only model updates with a central aggregation server. This decentralized training mechanism ensures that raw contractual data remains within the local environment, thereby preserving confidentiality while enabling collaborative learning across institutions.

The server sends the current global model parameters to all clients that are participating in each communication round. After that, each client trains its own model using stochastic gradient-based optimization, and then uses its own private dataset to modify the model parameters. The new parameters are then sent to the central server, where they are aggregated to generate a new global model. The Federated Averaging algorithm performs the aggregation step by calculating a weighted average of client parameters based on the size of their datasets. This iterative process continues across multiple communication rounds until model convergence is achieved across all participating clients.

The resulting hybrid architecture allows organizations to collaborate to develop models for understanding legal documents more accurately without exposing sensitive contract data at risk of harm. A series of experiments evaluate clause classification performance, the quality of generative explanations and the computational efficiency of the proposed setup across different system configurations to determine how effectively it performs. The following section presents the experimental setup and evaluation results.

IV. PERFORMANCE ANALYSIS

The effectiveness of the proposed hybrid federated setup was evaluated through a set of controlled experiments focusing on clause classification performance, generative summarization quality, computational efficiency, and privacy-preserving behavior under decentralized training. The experiments were conducted using the LEDGAR dataset, which consists of approximately 60,000 contractual clauses annotated across 98 semantic categories.

To simulate realistic deployment conditions in which legal data is distributed across independent organizations, the processed dataset was partitioned into three client subsets following a non-independent and non-identically distributed (non-IID) configuration. Each client received a distinct subset of clauses with variations in label distribution and clause frequency, reflecting heterogeneity typically observed across legal repositories. Local datasets were used exclusively for on-device training, ensuring that no raw contractual data was exchanged during the learning process, consistent with federated learning principles allowing clients to execute transparent data exchange [5].

Federated training was performed over multiple communication rounds. Each client updated model parameters locally using its private dataset and shared only model weight updates with the central server. The server aggregated client updates using the Federated Averaging (FedAvg) algorithm, producing a global model that captures knowledge across

distributed data sources. For the generative component, LoRA-based adapter training was employed, where only adapter parameters were updated and aggregated, while the base Phi-2 model remained frozen, reducing both memory usage and communication overhead. The evaluation process was primarily focused on two analytical tasks: clause classification and clause-level summarization. The performance of the centralized LegalBERT model was compared to that of the federated hybrid architecture for clause classification using identical data conditions. Table 1 shows a summary of the classification metrics that were observed in these experiments.

TABLE 1 Classification Performance of the proposed model on Top-20 Labels of LEDGAR

Model	Accuracy	Precision	Recall	F1 Score
Centralized LegalBERT	0.8800	0.7895	0.8224	0.7855
Federated Hybrid Model	0.8500	0.7406	0.7581	0.7287

Table 1 shows the accuracy, precision, recall and F1 score for both centralized and federated systems. Accuracy indicates the number of the clauses in the dataset were correctly classified. Precision evaluates the number of the predicted clause categories are correct. It is especially significant in legal document analysis. Incorrect classifications can lead to improper interpretations of contracts. Recall indicates us how effectively the model detects relevant clause categories in the dataset. The F1 score is used to illustrate the harmonic balance between precision-recall since the LEDGAR dataset has unbalanced clause distributions.

The centralized LegalBERT model has an accuracy of about 0.88, a precision of 0.79, a recall of 0.82, and an F1 score of 0.79. The federated hybrid architecture achieves slightly lower but comparable performance with roughly 0.85 accuracy, 0.74 precision, 0.76 recall, and 0.73 F1 score. The small reduction in classification performance observed in the federated configuration is expected since federated learning works on a distributed model. In federated environments, training data remains on each client's node, so the model cannot directly access the entire dataset during each optimization step [27]. Additionally, client datasets often have distinct and non-independent data distributions, which makes local training less stable. Despite these problems, the federated hybrid architecture still possesses a strong ability to make predictions while an additional advantage of ensuring sensitive contractual data remains protected in client environments. Comparative visualizations show the classification performance results by demonstrating how the metric distributions

Comparative visualizations in Fig.6 show the classification performance results by demonstrating how the metric distributions differ between different model configurations.

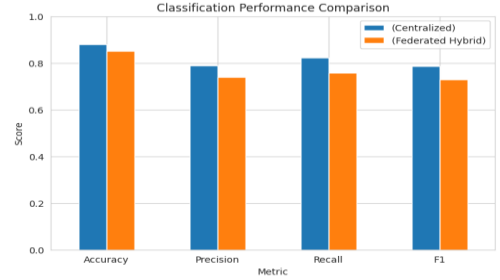


Fig. 6 Classification Performance Comparison between centralized and proposed hybrid dual-model

Additional figures including stacked metric comparisons, heatmap representations and radar charts in Fig.7 and Fig.8 further demonstrate how stable the proposed system is across multiple evaluation metrics.

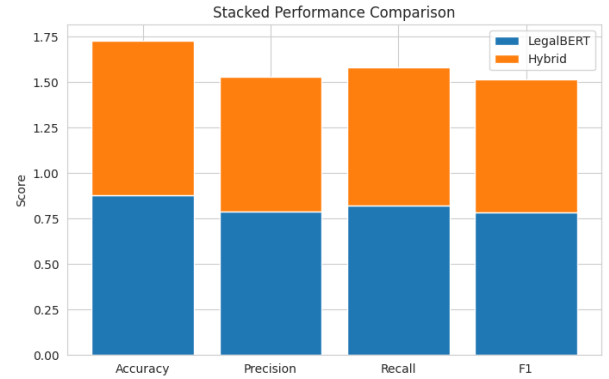


Fig. 7 Stacked Performance Comparison between centralized LegalBERT model and proposed hybrid model

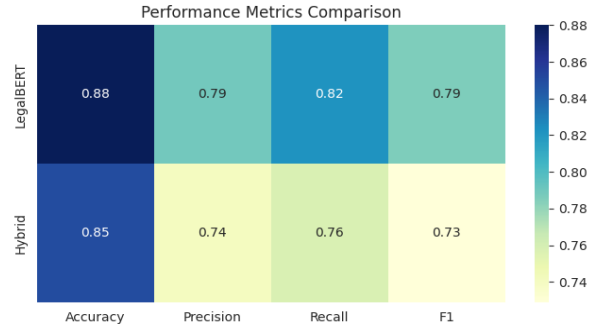


Fig. 8 Performance metrics heatmap comparison between centralized and proposed hybrid models

In addition to classifying clauses, the generative reasoning component of the architecture was evaluated using clause summarization experiments. The generative decoder model uses the Phi-2 language model with LoRA adapters trained in a federated environment to generate contextual explanations and summaries of contract clauses. The quality of generated summaries was evaluated using ROUGE metrics, which are widely used in natural language generation research to measure lexical and structural similarity between generated summaries and reference texts, assisting in the automatic evaluation of summaries generated by the decoder model [28].

The summarization performance results are summarized in Table 2.

TABLE 2 ROUGE Performance Metrics of the models evaluated on Top-20 Labels

Metric	Base Phi-2	Hybrid Federated Model
ROUGE-1	0.545	0.571
ROUGE-2	0.300	0.316
ROUGE-L	0.545	0.571

ROUGE-1 measures unigram overlap between generated and reference summaries, ROUGE-2 measures bigram overlap, and ROUGE-L evaluates the longest common subsequence similarity between generated and reference text sequences. The experimental results indicate the hybrid federated dual-model architecture achieves improved summarization performance compared to the base generative model. The hybrid model achieves ROUGE-1 of 0.57, ROUGE-2 of 0.32, and ROUGE-L of 0.57, which represent improvements over the base Phi-2 model without adapter fine-tuning.

The improvements in summarization quality can be attributed to the integration of clause classification information into the generative prompting process. Incorporating the predicted clause category to the generation prompt allows the language model more context, which helps it generate summaries. The comparative summarization performances using ROUGE scores is also illustrated through bar chart and radar chart images in Fig.9 and Fig.10 respectively.

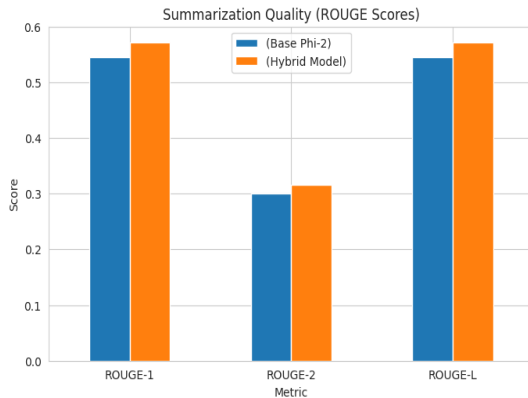


Fig. 9 ROUGE Scores Summarization Quality chart comparison between encoder model and proposed hybrid model

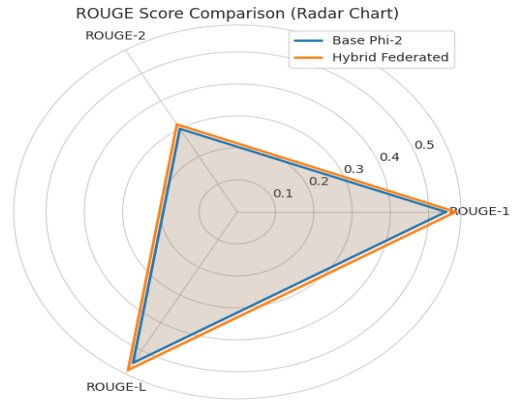


Fig. 10 ROUGE Score Comparison Radar plot between encoder model and proposed hybrid model

In addition to predictive accuracy and summarization quality, the system's computational efficiency was evaluated by measuring the duration it required to train different model configurations. Training time is a key component to consider when implementing real world deployments, where legal institutions might not be equipped with sufficient amounts of computing capacity. The experimental training time results are summarized in Table 3.

TABLE 3 Experimental Training times for all worked models

Model Configuration	Training Time (Minutes)
Centralized LegalBERT (encoder)	83.0
Federated LegalBERT (federated encoder)	44.8
Hybrid Federated System (Federated encoder + decoder)	47.3

The results in Table 3 show that centralized LegalBERT training required about 83 minutes, while federated LegalBERT training required only 44.8 minutes since the optimization was performed on the client side. The hybrid federated architecture which includes the Federated LegalBERT and the fine-tuned Phi-2 language model with LoRA adapters trained in a federated environment required 47.3 minutes since it needed additional computational resources for LoRA adapter training and hybrid inference integration. Despite this additional computation, the total training time is still considerably shorter than centralized training because the workload is distributed across the clients that are participating.

To further assess the overall effectiveness of the proposed architecture, classification and generative performance metrics were simultaneously evaluated across multiple system configurations, including the base generative model, centralized LegalBERT classifier, and the hybrid federated architecture. Table 4 shows a summary of the combined performance comparison.

TABLE 4 Overall Combined Performance across all models

Model	Clause Classification	Classification F1 Score	ROUGE-1
Base Phi-2 (General LLM decoder)	Prompt-based decoder	N/A	0.545
Centralized LegalBERT (encoder)	Supervised trained encoder	0.7855	N/A
Federated Hybrid Model	Federated + Supervised hybrid dual-model	0.7287	0.571

The results demonstrate that the base generative model has the ability to summarize information well, however it lacks a structured understanding of contractual clause categories. The centralized classification encoder model has fairly strong clause recognition, but it is unable to perform tasks such as contextual explanation or summarization. The hybrid federated architecture integrates structured clause classification with generative reasoning to ensure automated contract review achievable with both interpretability and contextual explanation. The overall model comparison is also illustrated through the bar chart below in Fig.11.

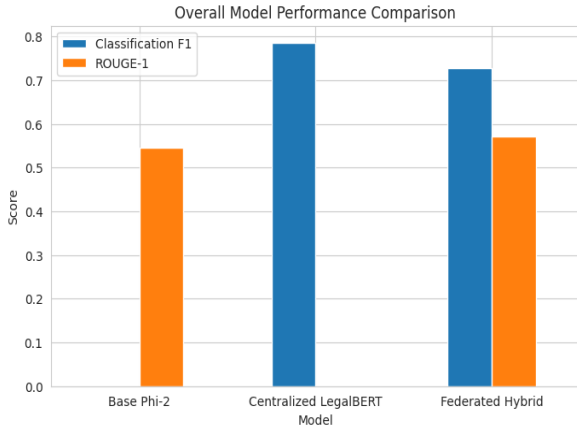


Fig. 11 Overall Models Performance Comparison bar chart

In addition to evaluating predictive performance, the proposed system was analysed from a human privacy and security perspective manually. Traditional centralized machine learning pipelines require contractual documents to be uploaded and stored within a central repository, which significantly increases the risk of unauthorized data access or large scale information leakage. In contrast, the federated learning architecture used in this work ensures that raw contractual data remains stored locally within each participating client environment.

Under the federated training structure, only model parameter updates are transmitted between clients and the central aggregation server. These updates include classification model parameter updates, and LoRA adapter parameter updates for the generative model. Because the original contractual documents are never transmitted outside their local environments, the system a lot reduces the risk of direct exposure of sensitive legal information.

Overall, the performance analysis proves that the proposed hybrid federated architecture provides a balance between predictive performance, computational efficiency and data privacy. While the federated configuration introduces minor reductions in classification accuracy due to distributed optimization constraints, it enables collaborative learning across organizations without requiring centralized collection of sensitive legal contracts. These results highlight the feasibility of applying federated large language models for real world legal contract review systems operating in privacy sensitive environments.

V. CONCLUSION

This work suggests a hybrid federated setup for analysing legal contracts while maintaining privacy by using LegalBERT for clause classification and a Phi-2 model with LoRA to explain and summarize them. The system is trained in a non-IID federated environment, which enables clients to learn collaboratively without sharing sensitive contract data. Results from experiments indicate that the federated model achieves competitive performance, achieving an accuracy of 0.85 in contrast to 0.88 for centralized training, alongside effective summarization performance as evaluated using ROUGE metrics. The slight decline in accuracy is primarily due to data heterogeneity and distributed training conditions, which are inherent issues associated with federated learning. In general, the proposed method demonstrates that it's possible to achieve contract analysis is both scalable and accurate while maintaining data private in real-world legal environments.

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